

Effect of forest conversion on soil water infiltration in the Dabie mountainous area, China

Yun Zhu^a, Lei Sun^a, Abdul Hakim Jamshidi^a, Xia Liu^{a,*}, Yiling Zheng^a, Zhaofei Fan^b

^a Co-Innovation Center for Sustainable Forestry in Southern China, Jiangsu Province Key Laboratory of Soil and Water Conservation and Ecological Restoration, Nanjing Forestry University, Nanjing, China

^b College of Forestry, Wildlife and Environment, Auburn University, Auburn, AL 36849, USA

ARTICLE INFO

Keywords:

Tea inter-planting
Land use and management
Litter-root-soil properties
Soil water infiltration

ABSTRACT

Study region: The Dabie mountainous area, China.

Study focus: Soil infiltration is a hydrological process dramatically affected by land use/cover changes. Soil infiltration characteristics, litter, root traits, and soil properties were monitored and the contributions of litter, root traits and soil properties on soil infiltrability were evaluated under *Castanea mollissima* forest (CMF), *Castanea mollissima*-tea inter-planting forest (CMTF), *Camellia oleifera* forest (COF), and *Camellia oleifera*-tea inter-planting forest (COTF), and miscellaneous woody forest (MWF, as the control).

New hydrological insights for the region: Significant differences were found in soil water infiltration following the change from natural to economic forest ($p < 0.05$). Soil initial infiltration rate (IIR), average infiltration (AIR), and steady infiltration rate (SIR) decreased by 35.65–61.59 %, 55.98–76.17 %, and 46.34–69.46 %, respectively. Soil water infiltration was more susceptible to soil properties than litter characteristics or root traits. Clay content was the primary factor affecting soil water infiltration. However, land use and management may drastically change litter and root trait conditions, which regulate soil properties, especially clay content. Therefore, management of slopping economic forests should improve the condition of litter and root traits to reduce soil and water loss hazards.

1. Introduction

Soil water infiltration is essential in regulating soil eco-hydrological processes, such as the formation and distribution of slope runoff, soil water regulation, and soil erosion (Ebel and Moody, 2013; Ando et al., 2015). Soil infiltration characteristics, such as initial, steady state, and average infiltration rates, are good indicators of soil quality and health and for understanding soil hydrological processes and their effects (Wu et al., 2016a, 2016b; Vergani and Graf, 2016; Shukla et al., 2003). Average infiltration rate (AIR) can reflect the overall speed of infiltration; initial infiltration rate (IIR) usually affects the runoff process in the early stages of rainfall, while steady infiltration rate (SIR) reflects soil infiltration capacity in the later stages of rainfall (Huang et al., 2019; Fu et al., 2021).

Land use and management activities can change vegetation characteristics such as tree canopy, litter coverage, root distribution, and litter and root-induced soil properties (Fig. 1). The impact of vegetation on the soil infiltration process is through the influence of

* Corresponding author.

E-mail address: liuxia@njfu.edu.cn (X. Liu).

the aboveground part of vegetation on the dynamic characteristics of runoff and the influence of the underground part of plant roots on soil physicochemical properties (Helvey and Patric, 1965; Wang et al., 2018; Hu et al., 2009; Zhu et al., 2020). A shift from mixed to these mono-species dominant forests is a significant threat to the ecosystem with loss of biodiversity and soil degradation. Shift from native forests to other forest types has been documented to affect underlying soil structure and associated properties like soil bulk density, porosity, infiltration, and runoff (Broersma et al., 1995; Liu et al., 2020). Thus, the conversion of native forests to other land-use types causes severe soil erosion (Wang et al., 2020; Cameron et al., 1981).

Inter-planting is vital for controlling soil erosion by increasing aboveground and underground biodiversity (Sharma et al., 2017; Guo et al., 2019). Inter-planting increases plant diversity, which would change the redistribution of rainfall through the canopy, and reduce the impact and stripping effect of raindrops in surface soil (Wu et al., 2016a, 2016b; Dexter, 2004). Furthermore, inter-planting generates more aboveground and underground biomass and thus improves organic matter content and soil porosity, contributing to water infiltration (Devitt and Smith, 2002; Zhu et al., 2019; Ilstedt et al., 2007). Research has found that economic forests inter-planted with perennial shrubs increased the number of fine roots and lateral roots. The roots' growth and decay can form macropores and more root channels, which are more conducive to water flow into the soil through channels (Wu et al., 2016a, 2016b; Bromley et al., 1997; Wu et al., 2021). In addition, plant roots influence soil infiltration by changing soil properties. Root systems play an essential role in inputting organic carbon and forming soil aggregates (Archer et al., 2002; Huang et al., 2017). On the one hand, high root biomass can increase soil water infiltration rate by increasing soil porosity and organic matter (Archer et al., 2002; Liu et al., 2019a, 2019b; Six et al., 2004); on the other hand, root systems may clog soil pores, resulting in a decrease in infiltration ability (Ng et al., 2013; Frouz et al., 2013).

However, Ilstedt et al. (2007) emphasized that it is inadequate to demonstrate the exact effects of inter-planting on improving soil water infiltrability. For example, in some cases, the degree to which inter-planting increases infiltration is much lower than expected. For example, volcanic soils may have higher inherent macroporosity, and tree roots and dead leaves have no significant impact, nor do they contribute to soil permeability. Therefore, understory vegetation may positively or negatively affect infiltration, depending on whether they are deeply rooted, perennial, or compacted and their interaction with the overstory trees. Understanding how inter-planting practices affect soil water infiltration under different sloping economic forests is limited.

Sloping economic forest lands are one of the primary sources of soil erosion in mountainous areas of China. These forests are widely distributed in mountainous and hilly regions of China, such as the Dabie mountainous area (PCSWES 2010; NSW CZ 2016). Occupying about 30.8 % of the total area of this region, soil and water loss reached 3492.83 km². Large-scale agricultural and forestry development activities have occurred in these forests since the 1980s. In the Dabie mountainous area, the economic forests occupy an area of over 30000 hm² and are dominated by *Castanea mollissima* and *Camellia oleifera* plantations. Approximately 80 % of these forests are distributed on sloping land.

This study aimed to examine soil infiltration characteristics of four dominant sloping economic forests, including *Castanea mollissima* (CMF), *Castanea mollissima*-tea inter-planting (CMTF), *Camellia oleifera* (COF), *Camellia oleifera*-tea inter-planting (COTF) economic forests in the Dabie mountainous area. Taking the miscellaneous woody shrubby and forest (MWF) as the reference (control), we intended to compare the differences in soil infiltration characteristics between these forests and potential driving factors. We hypothesized that (i) tea inter-planting management modes could enhance soil water infiltration characteristics compared to mono-cultivation of economic forest and (ii) soil properties had the highest contribution to infiltration characteristics compared to litter

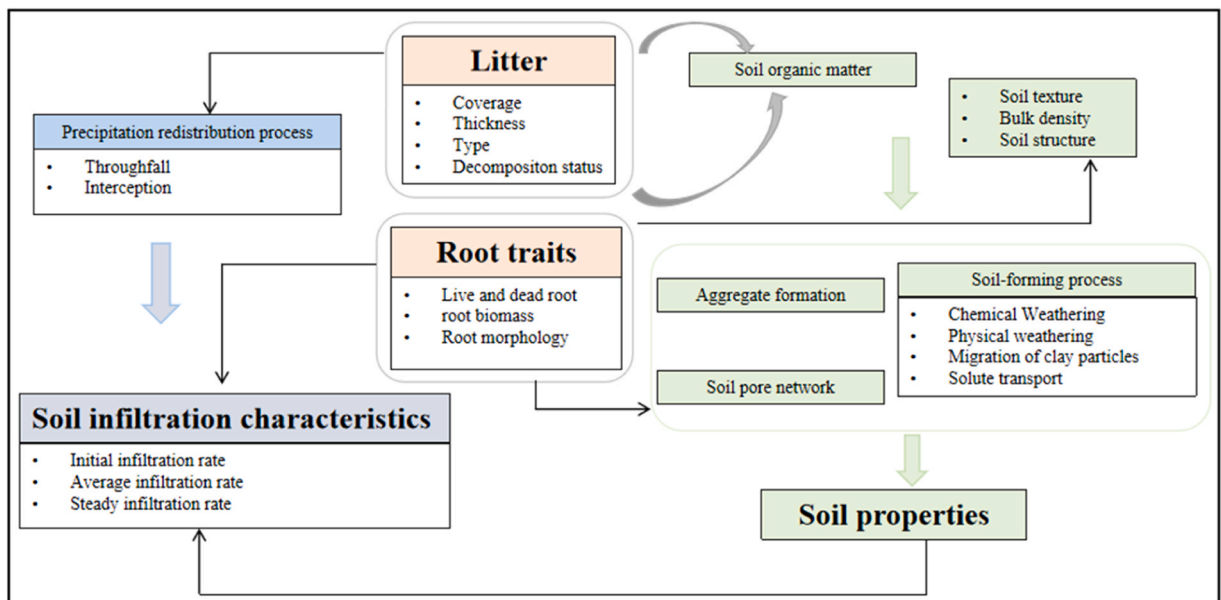


Fig. 1. A conceptual diagram of the effect of litter-root-soil interactions on soil water infiltration.

and root traits. Our results aimed to explore the key hydrological processes and mechanisms driving soil erosion after the transformation of economic forests to economic forests, to provide a reference for selecting the optimal soil cultivation mode and ecological restoration in the Dabie Mountain area.

2. Materials and methods

2.1. Study area

The study area is in Jinzhai County (Fig. 2), Anhui Province, in the north of the Dabie mountainous area ($115^{\circ}22'19''$ – $116^{\circ}11'52''$ E and $31^{\circ}06'41''$ – $31^{\circ}48'51''$ N), with a total area of 3814 km². This area belongs to the north subtropical humid monsoon climate, with four distinct seasons and abundant rainfall. Rainfall is mainly concentrated in summer. The annual average rainfall is about 1300 mm, accounting for 40–50 %. The annual average temperature is 15.3°C, and the terrain inclines from southwest to northeast. The highest point is Tiantangzhai on the southern border, with an altitude of 1729.13 m, and the lowest point is Guanji Village in Baitafan Town in the northeast, with an altitude of 60 m and a relative elevation difference of 1669.13 m.

At present, single-species economic forests and their inter-planting with tea mainly were developed from miscellaneous woody shrubby (MWF) since the late 1980s. The economic forest area accounts for 45.92 % of the total county area, mainly including *Castanea mollissima*, *Camellia oleifera*, *Carya cathayensis*, and *Vernicia fordii*, with *Castanea mollissima* and *Camellia oleifera* accounting for 60.15 % (Fig. 3), and the areas of *Castanea mollissima* and *Camellia oleifera* are 30,000 km² and 15,666 km², respectively. The *Castanea mollissima* and *Camellia oleifera* are mainly distributed in Baitafan Town, Meishan Town, and Huaishuwan Town.

2.2. Soil, litter, and root sampling

Based on the conceptual diagram (Fig. 1), a field survey on the distribution and site conditions of economic forests in Baitafan Town

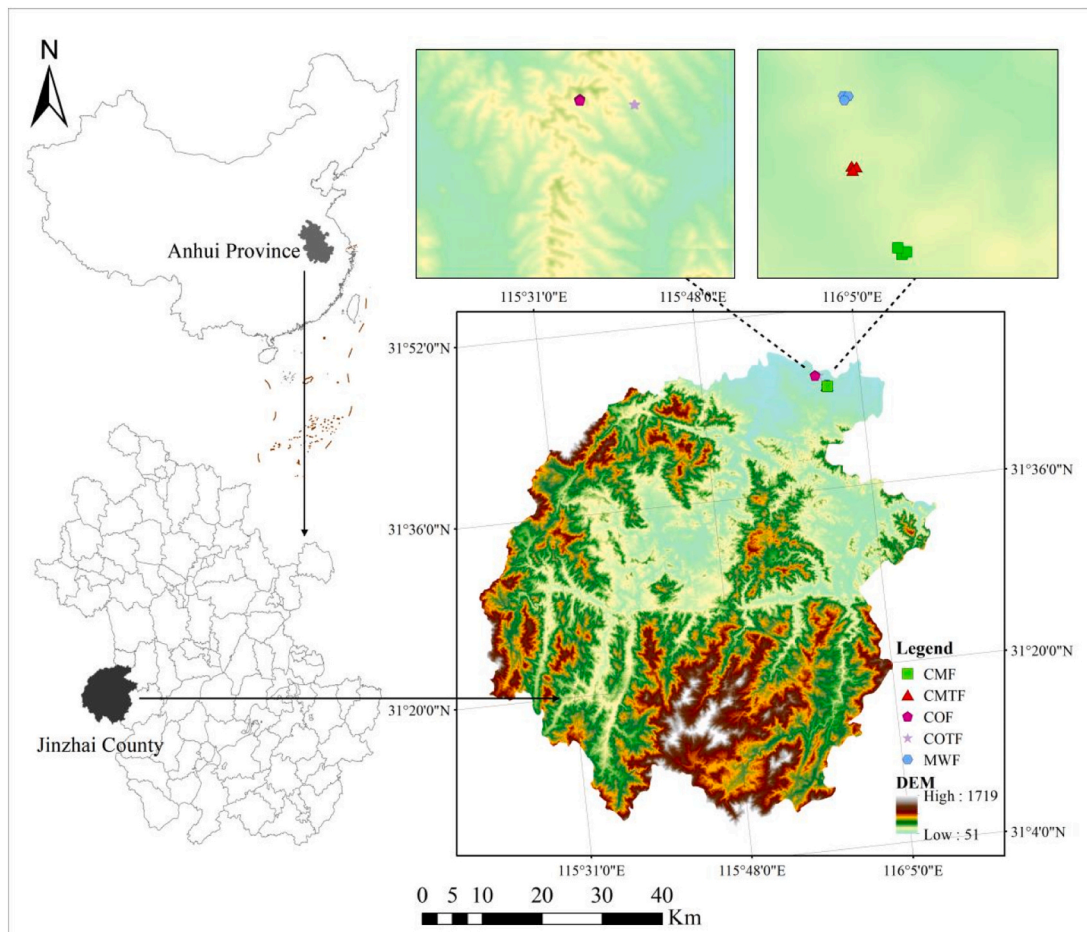


Fig. 2. The geographic position of the study sites. MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia Oleifera* forest; COTF, *Camellia Oleifera*-tea inter-planting.

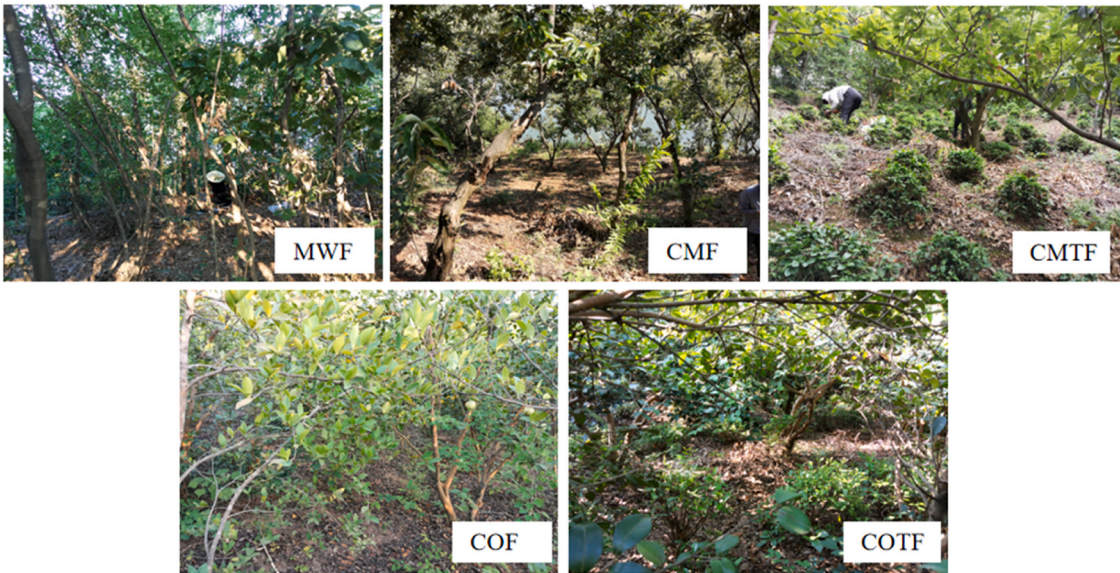


Fig. 3. Sites with different economic forests. MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia oleifera* forest; COTF, *Camellia oleifera*-tea inter-planting.

of Jinzhai County was conducted, and a total of 15 field sites with similar site conditions were selected in July 2022 (5 treatments $\times$ 3 replicates; Fig. 2), involving two different economic tree species (Fig. 3): *Castanea mollissima* (CMF) and *Camellia oleifera* (COF), and two different land management under each tree species: single planting and inter-planting with tea, *Castanea mollissima*-tea inter-planting forest (CMTF), and *Camellia oleifera*-tea inter-planting forest (COTF), respectively. Meanwhile, a miscellaneous woody shrubby as the control (MWF). In each field site, three sampling plots (10 m $\times$ 10 m) were set around the middle position of the slope (a total of 45 sampling plots). Vegetation species, average tree height (TH), stand density, row and plant spacing (RPS), average diameter at breast height (DBH), crown width, canopy density, soil thickness, litter thickness, and litter coverage in the plot were recorded in Table 1. In addition, since the research area is a typical rocky mountainous area with shallow soil layers, the cultivation of economic forests has a significant impact on the surface soil layer. Preliminary investigations had shown that water infiltration was concentrated in the surface layer (Fig. 4a). Meanwhile, the most active roots and root biomass were also distributed in the surface area (Fig. 4b). Thus, the tested soil layer is 0–20 cm (Niu et al., 2023; Liu et al., 2015).

We collected 0–20 cm cutting ring samples (100 cm³) from each plot to determine soil bulk density (BD), capillary porosity (CP), non-capillary porosity (NCP), and total porosity (TP). Five samples (in an ‘S’ shape) were collected from 0 to 20 cm to make a mixed soil sample and stored in plastic bags, which were then brought back to the laboratory for air-dried and passed through a 2 mm sieve to distinguish gravel (>2 mm) from fine soil (<2 mm). Soil gravel content (%) and essential soil physicochemical properties were determined. Besides, we set 3 sampling plots (1 m $\times$ 1 m) for litter thickness measurement. All the litter within the plots was collected and brought back to the laboratory for litter character determination. Soil samples for root determination were collected with a soil auger (with a diameter of 4 cm), and all the roots were carefully removed from the soil. A total of 90 cutting ring samples were collected from all plots (45 for measuring porosity and 45 for measuring infiltration characteristics), 45 bags of mixed soil samples, and 45 bags of litter.

Table 1
The basic information of the sampling plots.

T	SG	SD	DBH	TH	CW	ST	CD	VC	LC	LT	Age
MWF	8–15	27	/	/	/	/	95	15	80	35.55	/
CMF	8–15	21	22.3	6	7.5 $\times$ 6.5	3 $\times$ 3.5	65	10	70	23.76	20–30
CMTF	8–15	23	22.3	4.5	3 $\times$ 3.4	4 $\times$ 4	85	10	75	28.12	20–30 (CMF)
				0.5	0.5 $\times$ 0.5	2 $\times$ 2					5–10 (Tea)
COF	8–15	22	15.9	2.5	4 $\times$ 4	3 $\times$ 4	80	95	75	20.04	20–30
COTF	8–15	23	11.1	2.5	3.5 $\times$ 2.5	3 $\times$ 3	85	15	90	28.91	20–30 (COF)
				0.74	0.8 $\times$ 0.9	1 $\times$ 2					5–10 (Tea)

Notes: T, treatments; SG, slope gradient, $^{\circ}$; SD, soil depth, cm; DBH, diameter at breast height, cm; TH, tree height, cm; CW, crown width, cm; ST, space between trees, cm; CD, canopy density, %; VC, vegetation coverage, %; LC, litter coverage, %; LT, litter thickness, mm; MWF, Natural woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia Oleifera* forest; COTF, *Camellia Oleifera*-tea inter-planting.

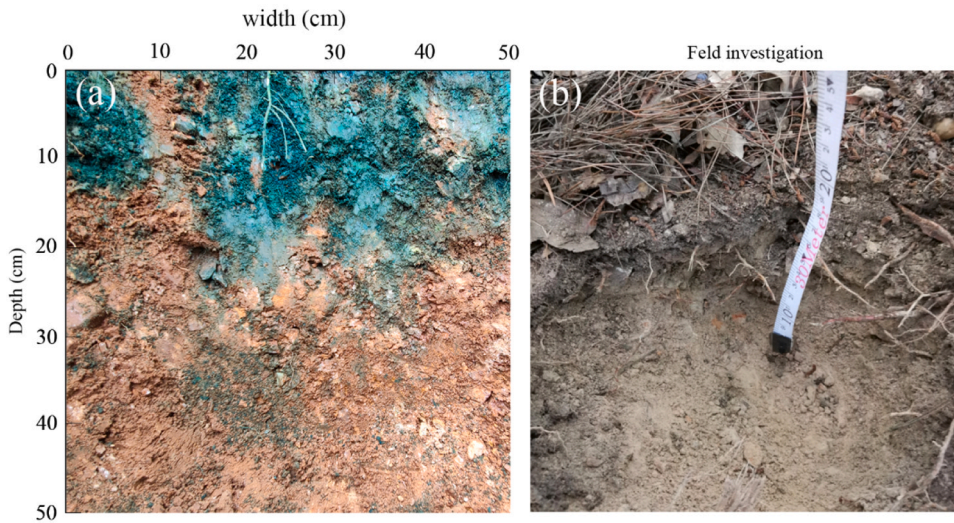


Fig. 4. Preliminary field investigation of soil (a) and root systems (b).

2.3. Determination of soil water infiltration characteristics

We used the single-ring infiltration method to measure the soil water infiltration characteristics due to the steep and uneven slope of the studied area which makes the double-ring method not applicable (Xu et al., 2012). Three experimental observation points were set in each sampling plot. For each plot, we had three replications where inside each replication we repeated the infiltration measurements by single ring in five different points. The locations with moderate slopes in each plot were selected for measurement and we retained the litter layers above the soil surface while measuring the infiltration. We kept the cross-section of the infiltration ring horizontal while inserting 5 cm of the ring into the soil. The amount of water infiltrated into the soil was monitored by filling the ring with water up to 7 cm and then recording the water level fall by time. We refilled back the water level to 7 cm anytime it was dropping to 3 cm meaning that we kept the ponding height varying between 7 and 3 cm.

The soil infiltration characteristics were described using three indicators: initial infiltration rate (IIR , $\text{cm} \cdot \text{min}^{-1}$), stable infiltration rate (SIR , $\text{cm} \cdot \text{min}^{-1}$), and average infiltration rate (AIR , $\text{cm} \cdot \text{min}^{-1}$). Among them, the IIR refers to the infiltration rate within the first 5 minutes; The SIR is the infiltration rate at which the infiltration amount tends to stabilize per unit time; The AIR is the average rate of the entire infiltration process, which is the ratio of the total infiltration amount to the infiltration time (Green and Ampt, 1911). In this experiment, the permeability rate of all undisturbed soil samples was stable within 80 minutes.

2.4. Laboratory analyses

2.4.1. Determination of water-holding characteristics of litter

The water-holding capacity of the litter was evaluated through the indoor soaking method (Chen et al., 2018; Li et al., 2015). The litter mass (m_1) was measured after oven-drying at 75°C (Li et al., 2022). The water content of litter soaked for 24 hours was regarded as the maximum water-holding capacity (R_m) and calculated as follows (Li et al., 2015; Zhou et al., 2018):

$$R_j = (m_j - m_1) / m_1 \times 100\% \quad (1)$$

where R_j is the water-holding capacity of the litter soaked for 5 min, 15 min, 30 min, 1 h, 2 h, 4 h, 8 h, 12 h, 24 h, and m_i is the litter mass after free drainage.

$$R_0 = (m_2 - m_1) / m_1 \times 100\% \quad (2)$$

Where m_2 is the fresh weight of litter.

$$R_m = (m_{24} - m_1) / m_1 \times 100\% \quad (3)$$

Where m_{24} is the litter mass soaked for 24 h after free drainage.

$$W_{eff} = (0.85R_m - R_0) m \quad (4)$$

$$W_{max} = (R_m - R_0) m \quad (5)$$

Where W_{eff} is the effective water-retention capacity (t ha^{-1}) and W_{max} is the maximum water-retention capacity (t ha^{-1}); m is the unit litter mass (t ha^{-1}).

2.4.2. Determination of root characteristics

Separated the roots from soil samples and washed them with water, followed by scanning and analyzing with an Epson V700 photo scanner (Tokyo, Japan) for trait parameters, including RLD (root length density) – defined as the total root length (mm) per 100 cm^3 , RSA (root surface area; cm^2) and RV (root volume; cm^3). Roots were then dried for 48 h at 75°C and weighed as RB (root biomass) (Liu et al., 2020).

2.4.3. Determination of soil physicochemical properties

Soil organic matter was measured using the potassium dichromate wet combustion method (Zhang et al., 2012). The volumetric soil particle size distribution was determined by laser granulometry (Mastersizer 2000, Malvern Co., England). Capillary porosity (CP) and non-capillary porosity (NCP) were measured with the water suction method (Wang et al., 2008). The CP (%), NCP (%), TP (%) and soil bulk density (BD) were calculated as follows:

$$CP = \frac{w_c \times BD}{V_t \times 100} \quad (6)$$

$$TP = \left(1 - \frac{BD}{D_s}\right) \times 100 \quad (7)$$

$$NCP = TP - CP \quad (8)$$

$$BD = \frac{m_{dry}}{V_t} \quad (9)$$

Where m_{dry} is the mass of dry soil, $g\ cm^{-3}$; V_t is the volume of the soil core, cm^{-3} ; w_c is the soil capillary water content, %; D_s is the soil particle density, $2.65\ g\ cm^{-3}$.

2.5. Statistical analysis

One-way analysis of variance (ANOVA) was conducted to test the difference of soil water infiltration characteristics, litter, root traits, and soil properties among different land use types, followed by Duncan post-hoc tests. Pearson's correlation analysis was used to explore the relationships among litter, root traits, and soil properties. The influence of litter characteristics, root traits, and soil properties variables upon soil infiltration characteristics was evaluated by variance partitioning analysis (VPA), using the vegan package for R.4.3.1 (Dixon., 2003; Sanchez, 2013).

The Canoco 5.0 software was used for Redundancy Analysis (RDA). We established a matrix of species data and environmental data in RDA. The species data are IIR, AIR, and SIR; environmental data included LC, LT, R_m , W_{eff} , W_{max} , RV, RD, RS, RB, Clay, Sand, Silt, Gravel, CP, NCP, TP, SOM and BD. The Structural Equation Model (SEM) is mainly used to analyze the direct and indirect effects between LC, LT, R_m , W_{eff} , W_{max} , RV, RD, RS, RB, Clay, Sand, Silt, Gravel, CP, NCP, TP, SOM, BD, and their impact on IIR, AIR, and SIR. An initial path network was constructed using AMOS 24.0 software to explain the reasons for variable changes. The standardized path coefficients were estimated by the maximum likelihood method. Model evaluation adopted χ^2 . The Comparative Fit Index (CFI) and Root Mean Square Error of Approximation (RMSEA) were tested and compared, and the correction index was used to correct the model to improve its fit (Wei et al., 2018).

3. Results

3.1. Soil water infiltration characteristics

The soil infiltration rates, regardless of land use and management, decreased sharply within the first 10 mins and then tended to be stable, following robust power functions ($R^2 = 0.88 - 0.98$, Fig. 5). The AIR, IIR, and SIR decreased 35.65–61.59 %, 55.98–76.17 % and 46.34–69.46 % significantly after the native forests were developed to economic forests ($p < 0.05$; Fig. 6). In addition, the soil

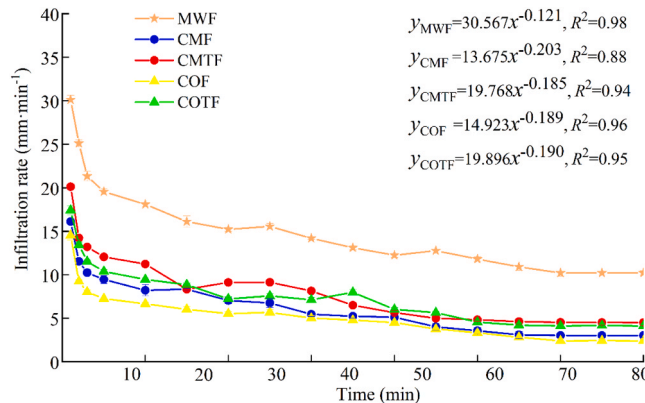


Fig. 5. Change of infiltration rate (mean $\pm$ SE, $n = 3$) with time under different land use and management types. MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia oleifera* forest; COTF, *Camellia oleifera*-tea inter-planting.

infiltration characteristics increased significantly after adopting tea-inter-planting on mono-cultivated economic forest. The CMTF had higher IIR and SIR than COTF ($p < 0.05$; Fig. 6a and c), but no significant difference was found in AIR between CMTF and COTF ($p > 0.05$; Fig. 6b).

3.2. Litter characteristics

At the beginning of the experiment, the water absorption rate was greatest and declined rapidly in the first 1 h, gradually slowed down and stabilized after 12 h, and approached 0 until 24 h (Fig. 7a). The litter thickness, litter coverage, R_m , W_{eff} and W_{max} significantly differed among different land use types of sloping economic forests, which decreased 23.18–47.89 %, 20.31–52.94 %, 43.31–59.59 %, 23.75–78.07 % and 22.72–80.08 % after the native forest was developed into an economic forest (Fig. 7b, c, d, e and f; $p < 0.05$). The litter thickness, W_{eff} , and W_{max} were higher in CMF than in COF (Fig. 7b, e, and f), while litter coverage was lower than in COF (Fig. 7c). No significant difference in R_m between CMF and COF (Fig. 7d; $p > 0.05$). After implementing tea-inter-planting measures on the mono-cultivated economic forest, the litter coverage, thickness, R_m , W_{eff} , and W_{max} increased significantly. Meanwhile, W_{eff} and W_{max} were higher in COMF than in COTF (Fig. 7e, and f), but COTF had higher litter coverage (Fig. 7c; $p > 0.05$), no significant difference was found between COMF and COTF in litter thickness and R_m (Fig. 7b and d; $p > 0.05$). In addition, the MWF had the maximum interception capacity, which was equivalent to 6.02 mm of rainfall (note that 1 mm of precipitation is equivalent to 1 t ha⁻¹), with 2.25 mm in the CMF, 1.32 mm in the COF, 4.51 mm in the CMTF and 3.01 mm in the COTF (Fig. 7e).

3.3. Root traits

Fig. 8 shows the distribution of root systems in the soil. As a whole, the root characteristics follow the trends MWF > CMTF > COTF > CMF > COF. The RLD, RSA, RV, and RB decreased significantly after native forests were developed into economic forests. The ranges are 14.6–85.4 %, 23.0–78.8 %, 34.9–75.9 %, and 46.3–84.4 % (Fig. 9). Meanwhile, the CMF had higher RLD, RSA, RV, and RD than COF (Fig. 9; $p < 0.05$). Tea-interplanting increased RLD, RSA, RV, and RD compared with mono-cultivated economic forest. The RLD, RSA, RV, and RD were higher in CMTF than in COTF (Fig. 9; $p < 0.05$).

3.4. Soil physicochemical properties

Overall, soil physicochemical properties significantly differed by land use and management types. Soil organic matter, CP, NCP, TP, and the silt and clay content decreased significantly after converting native forests to economic forests. In contrast, BD, gravel, and sand content were increased (Table 2). The CMF had higher SOM, gravel, CP, NCP, TP, and sand content than COF but lower BD, silt, and clay content ($p < 0.05$). In addition, tea-inter-planting forests had higher SOM, CP, NCP, TP, and silt and clay content than mono-cultivated economic forests but had lower BD, gravel, and sand content. In CMTF, the BD, silt, and clay content were lower than COTF, and the gravel and sand content were higher, while no significant differences were found in SOM and soil porosity between them ($p > 0.05$).

Correlation analysis indicated that RD, RSA, RV, and RB had a significantly positive correlation with SOM, CP, NCP, TP, silt, and clay content but negatively correlated with BD, sand, and gravel content (Fig. 10; $p < 0.01$). No significant correlation was found between RD, RSA, and silt content, while silt content had a significant positive correlation with RV and RB ($p > 0.05$). The LT, LC, R_m , W_{eff} , and W_{max} had a negative correlation with BD, sand, and gravel content, while a positive correlation with SOM, CP, NCP, TP, clay, and silt content, except no significant correlation was found between W_{eff} , W_{max} , and silt content (Fig. 10; $p > 0.05$).

3.5. Relationships among soil water infiltration and litter, root, soil

The VPA analysis showed that selected environmental variables could explain 99.99 % of the differences in soil water infiltration

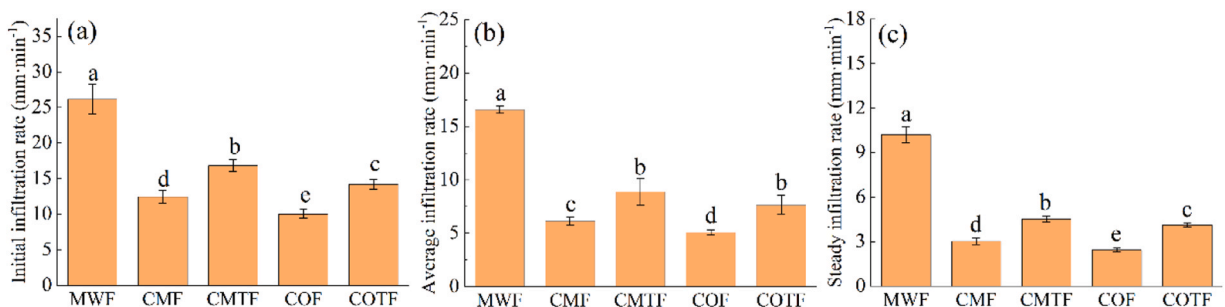


Fig. 6. Soil initial infiltration rate (a), average infiltration rate (b) and steady state infiltration rate (c) under different land use and management types. Different lowercase letters in each sub-graph indicate significant differences among treatments at 0.05 levels. MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia oleifera* forest; COTF, *Camellia oleifera*-tea inter-planting.

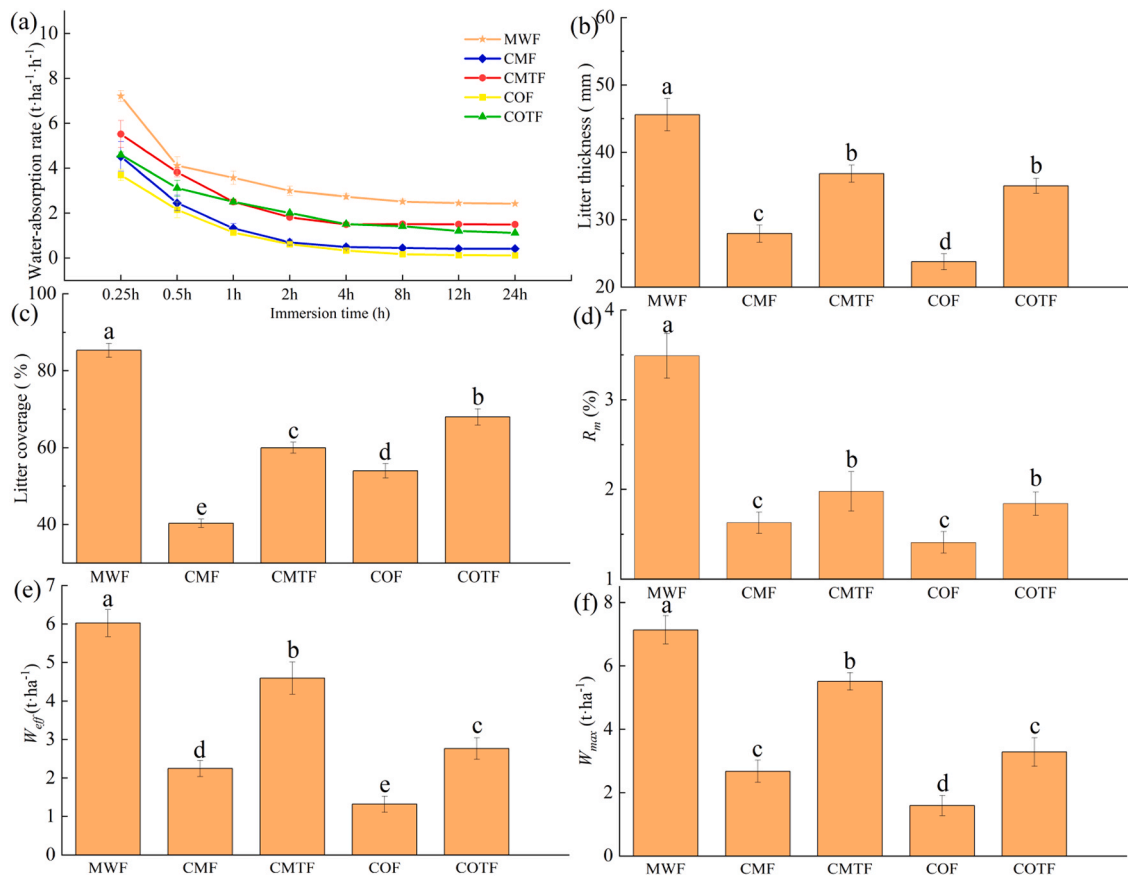


Fig. 7. Variations in the water absorption rate (a), Litter thickness (b), Litter coverage (c), R_m (d), W_{eff} (e) and W_{max} (f) under different land use and management types. Different lowercase letters in each sub-graph indicate significant difference among treatments at 0.05 level. R_m , maximum water-holding capacity of the litter; W_{eff} , effective water-retention capacity; W_{max} , maximum water-retention capacity; MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia Oleifera* forest; COTF, *Camellia Oleifera*-tea inter-planting.

characteristics (IIR , AIR , and SIR). The comprehensive effects of litter characteristics, root traits, and soil properties explained 79.06 % of the total differences. Soil properties contributed more to soil water infiltration than litter characteristics and root traits. The contribution of the interaction effect between litter characteristics and root traits to the total difference is 10.81 %, much higher than other interaction effects (Fig. 11).

The RDA results indicated that the first and second sorting axes explain 94.56 % and 4.32 % of the total variance, respectively, with a cumulative explanation of 98.88 % for the first two axes (Fig. 12). The first and second axes can fully reflect the relationship between soil water infiltration characteristics and litter, root traits, soil properties, and are mainly determined by the 1 axis. The partial Monte Carlo Permutation test showed that the clay content, which accounts for 86.23 %, has the highest degree of interpretation, which is the main factor affecting soil water infiltration (Table 3). Soil water infiltration characteristics were positively correlated with R_m , RD, RSA, RV, RB, LC, LT, W_{eff} , TP, and clay content ($p < 0.05$), while were negatively correlated with BD, Sand, and Gravel content ($p < 0.05$).

The SEM was applied to explore the direct and indirect effects between the independent and dependent variables. The LC, RB, CP, TP, NCP, R_m , SOM, and clay content directly and significantly affected AIR , IIR , and SIR . Although land use types had no direct effect on AIR , IIR , and SIR , they still exhibited evident total effects on AIR , IIR , and SIR (standardized path coefficient of -0.465 , -0.49 , -0.51). Land use types change from native forest to economic forest would decrease AIR by reducing LC to affect the R_m , and then a decrease in R_m indirectly decreased soil silt content and SOM, decreasing TP (Fig. 13a). Meanwhile, land use types decreased the R_m by decreasing LC. A decrease in RB indirectly affected soil clay content and SOM and then decreased CP, resulting in a decrease in IIR (Fig. 13b). However, unlike AIR , the SIR decreased by reducing CP (Fig. 13c).

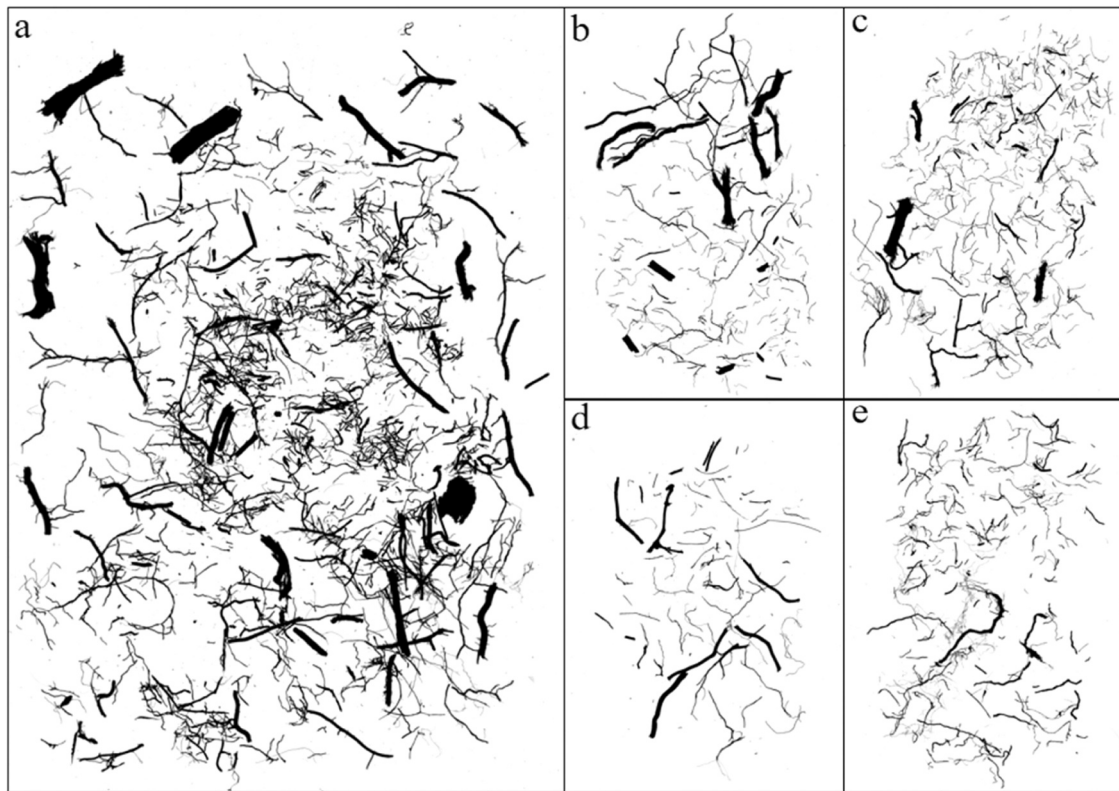


Fig. 8. The real images of root collected from 0 to 20 cm under MWF (a), CMF (b), CMTF (c), COF (d), and COTF (e). MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia Oleifera* forest; COTF, *Camellia Oleifera*-tea inter-planting.

4. Discussion

4.1. Effect of land use change on soil water infiltration characteristics

The soil infiltration characteristics had significant differences among different land use types, which has been reported by other studies in different regions with various land use types (Yimer et al., 2008; Gonzalez-Sosa et al., 2010; Neris et al., 2012; Huang et al., 2015). The changes in land use systems, mainly through the conversion of natural vegetation significantly influence various natural environmental processes. In the present study, the MWF had the highest soil infiltration rate, IIR, AIR, and SIR compared with CMF and COF (Figs. 5 and 6).

Differences in the infiltration characteristics among different land use types should be owed to the relevant environmental factors and human disturbances, the variation of vegetation coverage, litterfall, root changes, and soil properties would lead to corresponding differences in soil pores, particle composition, and SOM, et al. (Neris et al., 2012; Poulenard et al., 2001; Mo et al., 2005). Our result showed that soil water infiltration characteristics were positively correlated with RD, RSA, RV, RB, LC, and LT (Fig. 11). After MWF was changed to CMF and COF, the loss of vegetation and litter caused the soil surface to be washed away by rainwater and stripped by runoff, causing the reduction of soil water infiltration, leading to soil erosion (Jiang et al., 2019; Rodríguez Rodríguez et al., 2002). The impact of the litter layer on infiltration is mainly reflected in intercepting rainfall in the vertical position and blocking slope flow in the horizontal position. Meanwhile, the incorporation of litter into surface soil promotes aggregate formation and stability and increases soil surface roughness and porosity (Zhang and Xie., 2019; Sun et al., 2018; Zhao et al., 2014).

In addition, the reduction of root characteristics (RB, RD, RSA, RV) can cause soil particles to rearrange and secretion to decrease, leading to changes in soil properties, which in turn reduces the soil infiltration characteristics (Jørgensen et al., 2002; Bronick and Lal, 2005). The root is essential in decreasing runoff and holding soil (Pan et al., 2016; Sun et al., 2014). Root systems determine the spatial distribution of root channels in the soil and affect water infiltration. In addition, plant roots can increase soil porosity and organic matter in the soil to improve soil infiltration capacity (Leung et al., 2015; Huang et al., 2017). This is consistent with our results that root traits had a significant positive correlation with soil porosity and SOM (Fig. 10; $p < 0.05$). Therefore, our results indicated that those changes in land use types negatively affect soil water infiltration characteristics; the development of sloping economic forests increased soil runoff and led to soil erosion.

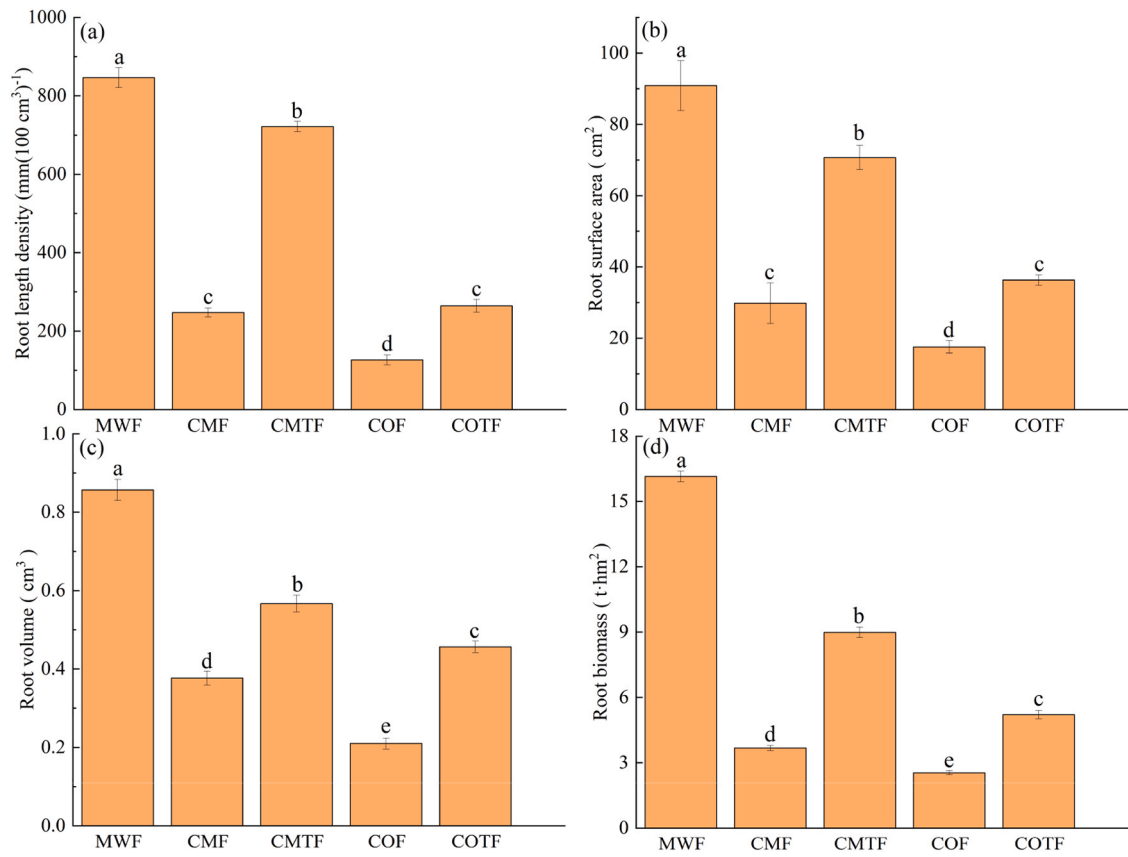


Fig. 9. Root length density (a), root surface area(b), root volume (c), and root biomass (d) under different land use and management types. Values are means $\pm$ SE (n = 3). Bars with different letters indicate significant differences at 0.05 levels. MWF, Miscellaneous woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia Oleifera* forest; COTF, *Camellia Oleifera*-tea inter-planting.

Table 2

Soil physicochemical properties (mean $\pm$ standard error) under different land use and management types (T).

T	BD	SOM	Gravel (%)	CP	NCP	TP	Soil mechanical composition (%)		
							Sand (2–0.05 mm)	Silt (0.05–0.002 mm)	Clay (<0.002 mm)
MWF	1.21 ± 0.03 d	44.80 ± 2.15 a	22.11 ± 2.49 e	23.87 ± 2.89 a	23.35 ± 1.81 a	47.22 ± 1.83 a	62.56 $\pm$ 4.07 d	35.97 $\pm$ 3.92 a	1.47 $\pm$ 0.17 a
CMF	1.50 ± 0.02 b	35.79 ± 1.36 c	48.53 ± 1.30 a	14.44 ± 1.69 c	14.05 ± 1.16 c	28.49 ± 1.57 d	80.31 $\pm$ 3.74 a	19.43 $\pm$ 3.75c	0.26 $\pm$ 0.04 d
CMTF	1.44 ± 0.01 c	39.09 ± 1.53 b	34.97 ± 1.39 c	19.79 ± 1.90 b	18.99 ± 1.27 b	38.78 ± 2.30 b	71.45 $\pm$ 0.63c	28.77 $\pm$ 0.59 b	0.68 $\pm$ 0.05c
COF	1.58 ± 0.02 a	26.23 ± 0.73 d	38.25 ± 1.33 b	11.61 ± 1.32 d	9.28 $\pm$ 1.27 d	20.89 ± 0.95 e	76.05 $\pm$ 1.56 b	23.37 $\pm$ 1.58c	0.58 $\pm$ 0.03c
COTF	1.48 ± 0.02 b	36.84 ± 1.95 bc	30.26 ± 1.27 d	17.45 ± 2.37 bc	16.45 ± 2.37 b	33.90 ± 1.89 c	66.43 $\pm$ 0.97 d	32.65 $\pm$ 1.06 a	0.92 $\pm$ 0.10 b

Different letters in same column indicate significant differences at 0.05 level.

MWF, Natural woody forest; CMF, *Castanea mollissima* forest; CMTF, *Castanea mollissima*-tea inter-planting; COF, *Camellia Oleifera* forest; COTF, *Camellia Oleifera*-tea inter-planting; BD, bulk density, g cm^{-3} ; SOM, soil organic matter content, $\text{g} \cdot \text{kg}^{-1}$; CP, capillary porosity, %; NCP, Non-capillary porosity, %; TP, total porosity, %.

4.2. Effect of land use management on soil water infiltration characteristics

Research has shown that tree species can impact soil water infiltration (Jačka et al., 2014; Kan et al., 2020). We found that the CMF had higher water infiltrability than COF (Figs. 5 and 6), accompanied by the greatest SOM, clay content, soil porosity, and lowest BD values in the topsoil of CMF (Table 2). These values correspond with the highest infiltration rate in CMF because of its more extensive canopy cover and roots. The large tree crown can effectively intercept rainwater, prevent raindrops from directly splashing on the

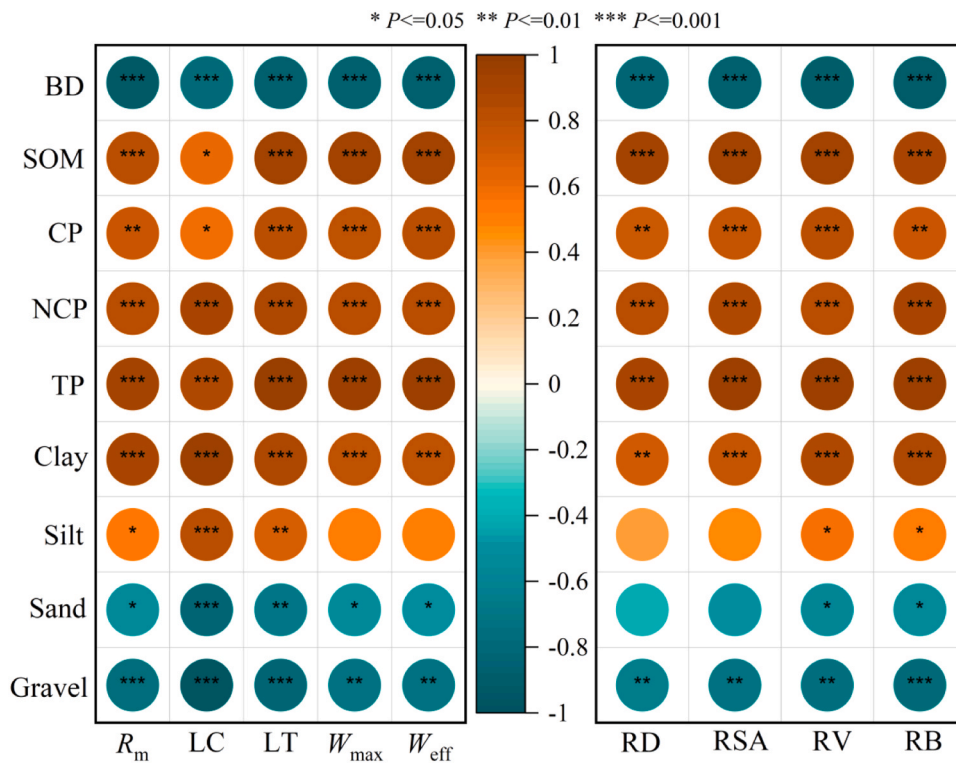


Fig. 10. Correlation coefficients between litter, root traits and soil properties. RD, root density; RSA, root surface area; RV, root volume; RB, root biomass. LT, litter thickness; LC, litter coverage; R_m , maximum water-holding capacity of the litter; W_{eff} , effective water-retention capacity; W_{max} , maximum water-retention capacity; BD, soil bulk density; NCP, non-capillary porosity; CP, capillary porosity; TP, total porosity; SOM, soil organic matter content; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

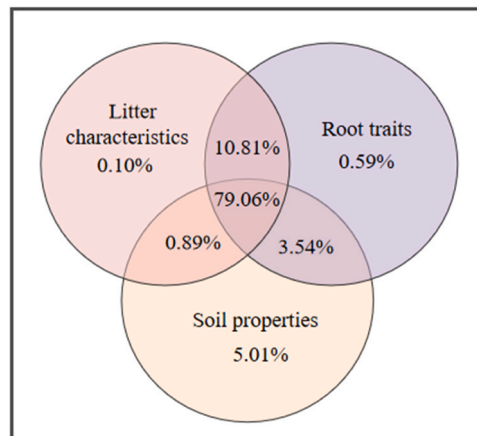


Fig. 11. The variation partitioning analysis (VPA) on the relative contribution of litter, root characteristics and soil properties to the soil water characteristics (IIR, AIR and SIR) under different land use and management types.

surface, and prevent surface soil damage conducive to soil water infiltration. However, the COF belongs to shrubs with smaller crowns than CMF, and the soil is more susceptible to rainwater erosion. In addition, the tree roots of CMF are large and deep, while COF roots are small, thin, shallow, and relatively low (Fig. 8). Recent studies have shown that plants with coarser roots mainly increased aeration pores and had higher porosity than other root plants (Liu et al., 2016; Tollner et al., 1990).

Studies have shown that inter-planting modes influence water infiltration (Molina et al., 2007; Burr-Hersey et al., 2020; Zhu et al., 2018). In our study, compared with monoculture economic forests, the degree of soil degradation in the tea inter-planting forests was lower, and the soil water infiltration abilities (e.g., IIR, AIR, and SIR) were higher (Figs. 5 and 6). Our results are consistent with Cannavo's et al. (2011) previous research. It may be the reason that soil properties improved because of the higher vegetation cover

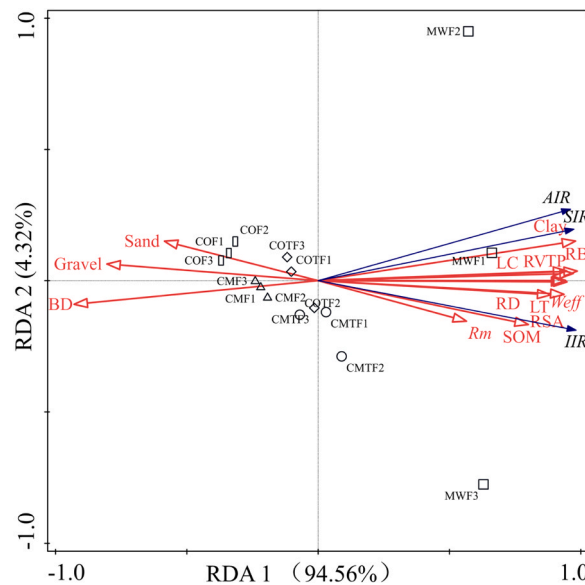


Fig. 12. The redundancy analysis (RDA) based on selected environmental variables and soil water infiltration characteristics. Values of Axes 1 and 2 are percent of variability explained by the corresponding axes. The length of the arrow indicates the relative variance explained by that factor. Red vectors represent litter characteristics (LC, litter coverage; LT, litter thickness; R_m , maximum water-holding capacity of the litter; W_{eff} , effective water-retention capacity; W_{max} , maximum water-retention capacity), root traits (RB, root biomass; RD, root density; RSA, root surface area; RV, root volume) and soil properties (NCP, non-capillary porosity; CP, capillary porosity; TP, total porosity; BD, soil bulk density; SOM, soil organic matter content); blue vectors represent soil water infiltration characteristics.

Table 3

The explained variability and significance of each environmental variable to soil water infiltration characteristics.

Environmental variables	Explains (%)	pseudo- <i>F</i>	<i>P</i>
Clay	86.23	80.34	0.002
RB	8.23	30.37	0.002
LC	2.39	2.45	0.022

LC, litter coverage; RB, root biomass.

and SOM, thicker litter layer, and more root systems (Tables 1 and 2). In the mechanism of inter-planting to control soil erosion, the above-ground and underground parts mainly play a role in the impact and regulation of inter-planting on soil erosion. From the above-ground perspective, tea trees with a thicker litter layer could prevent soil particle splashing and pore blockage, resulting in an increase in soil water infiltration (Zhu et al., 2018; Bargaúes Tobella et al., 2014). Furthermore, more litter residues reduced the water flow speed and promoted soil infiltration ability (Lehmann et al., 2001; Chandler and Chappell, 2008).

The root system also plays a vital role below ground in controlling soil erosion, such as changing the spatial distribution of roots and increasing root density and biomass. These changes in the root system alter water distribution during rainfall, increase the absorption and storage of soil moisture, increase soil moisture infiltration, and reduce surface runoff generation (Wu et al., 2016a, 2016b; Ghestem et al., 2014). Meanwhile, root decay and litter residues significantly increased the input of soil organic matter beneath the trees (Cong et al., 2023; Lacombe et al., 2009; Persson, 1980). In contrast, Benegas et al. (2014) found that inter-planting had no significant impact on soil infiltrability under the coffee agroforestry system in Central America, possibly because the soil on slopes was severely compacted by daily management activities, such as picking, trampling, soil turning, and loosening processes in our study. The topsoil layer with cultivation had higher soil bulk density, denser structure, and lower total porosity (Table 2; Liu et al., 2016), which caused a lower soil water infiltration (Blevins et al., 1970; Zhu et al., 2018; Whalley et al., 2004). These are the reasons why tea inter-planting increased soil infiltration and reduced erosion compared with the single development of a sloping economic forest.

4.3. The factors affecting soil water infiltration under different land use and management types

Litter characteristics, root traits, and soil properties were all found to be essential key factors affecting soil infiltrability (Cui et al., 2019; Shi et al., 2022; Wu et al., 2016a, 2016b). However, we found that their relative contributions were not equal. The VPA analyses showed that soil water infiltration was primarily determined by litter characteristics (0.07 %) and root traits (0.51 %) versus soil properties (0.81 %) with their combined effects (80.15 %). Soil properties contributed more to soil water infiltration than litter characteristics and root traits (Fig. 11). Litter and root traits indirectly affect soil water infiltration by influencing soil properties, such

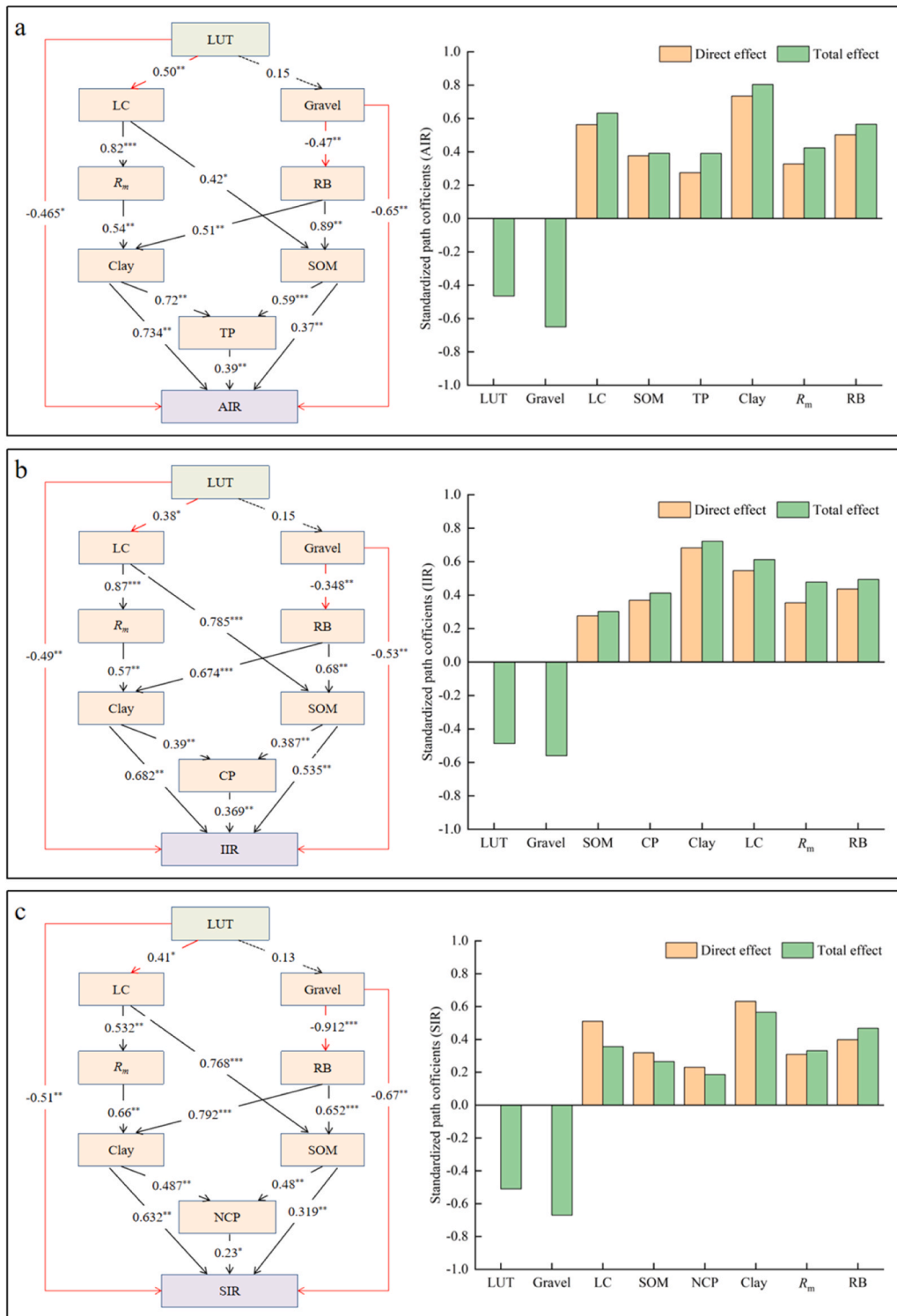


Fig. 13. Structure equation model (SEM) relating litter, root traits, soil properties and average infiltration rate (AIR; a), initial infiltration rate (IIR; b) and stable infiltration rate (SIR; c) under different land use and management types (LUT). Rectangles represent observed variables and arrow represent path coefficient. Numbers on arrows are standardized path coefficients; Black lines indicate positive effects; red lines indicate negative effects; dashed line indicates no significant difference; significance levels are: *, $p < 0.05$; **, $p < 0.01$ and ***, $p < 0.001$. The data near the lines represent the standardized regression weights; LC, litter coverage; R_m , maximum water-holding capacity of the litter; NCP, non-capillary porosity; CP, capillary porosity; TP, total porosity; RB, root biomass; SOM, soil organic matter.

as organic matter, porosity, and the like (Huang et al., 2017; Bargués Tobella et al., 2014; Jotisankasa and Sirirattanachai, 2017).

Furthermore, RDA analysis showed the clay content played a crucial role in determining soil infiltration (Table 3), and soil water infiltration was significantly positively correlated with the clay content (Fig. 12), which is the same result as Wang et al. (2023). The clay content has a positive effect on root traits, SOM, and soil porosity (Ali et al., 2017; Borden et al., 2020), which subsequently exerts a significant influence on soil infiltration ability (Archer et al., 2002; Jiang et al., 2018; Thompson et al., 2010). Human activities significantly impact soil particle composition under different land use types in our research area. After MWF was developed into CMF and COF, due to daily land management activities such as picking, trampling, and weeding, human activities have caused serious disturbances. The topsoil layer with cultivation had lower total porosity, denser structure, and the fine particles lost seriously (Table 2; Zhu et al., 2018), which have a significant negative effect on soil water infiltration (Whalley et al., 2004; Abdul-Megid et al., 1987). However, other studies showed that there was a significant negative correlation between clay content and soil infiltration rate; the higher the clay content, the smaller the soil pores and the greater the viscous resistance, resulting in a lower soil infiltration rate (Alletto and Coquet, 2009; Chartier et al., 2011; Wang et al., 2017), probably attributed to different soil textures in the study area, the Dabie Mountains area belongs to rock mountainous areas with a high content of sand (Table 2).

The SEM analysis showed that land use types affected R_m by influencing litter coverage and the root biomasses, subsequently affecting soil porosity by influencing the clay content, ultimately altering soil water infiltration (Fig. 13). The litter covers the ground, prolonging the contact time between water and soil, allowing water to penetrate the soil more evenly (Sun et al., 2018). Secondly, it increases SOM, improves soil structure, and enhances soil water holding capacity after decomposition (Chen et al., 2018; Tu et al., 2022). Meanwhile, high root biomass directly stimulates the increase of SOM and soil porosity, which conduces to higher soil water infiltration ability (Bronick and Lal, 2005; Galloway et al., 2018; Huang et al., 2017; Liu et al., 2019a, 2019b), coincident with the RB had positive correlation with SOM, CP, NCP, and TP (Fig. 9). In our study, the clay content had a significant positive effect on soil infiltrability by increasing soil porosity (Fig. 13), which is consistent with previous research (Liu et al., 2019a, 2019b). Moreover, soil porosity affects the continuity and size of the water flow path, and has a direct positive impact on soil infiltrability (Hao et al., 2020; Marquart et al., 2020), which is consistent with our result (Fig. 13). The study showed that in Dabie Mountain area with poor soil texture and structure, soil infiltration capacity can be improved by altering soil particle composition and pore structure, such as effectively increasing the fine-grained content substances such as clay and silt particles.

5. Conclusion

We evaluated the effect of litter, root system, and soil characteristics on soil water infiltration characteristics under different cultivation modes of sloping economic forests, and explored the main influencing factors. The changes in land use types from native forests to economic forests significantly reduced soil water infiltrability. The extent of reduction was different due to different economic forest species. The soil under *Castanea mollissima* had better water infiltrability than *Camellia oleifera*, regardless of the land management types. Tea inter-planting promotes soil water infiltration of economic forests due to the lower bulk density and higher clay content, porosity, SOM, root, and litter mass. Soil properties were the main contributor to soil water infiltration characteristics compared to litter and root traits, and the clay content was the main factor affecting soil infiltration characteristics. The clay content should be considered an indicator when evaluating soil infiltrability in the Dabie Mountains area.

In conclusion, implementing tea inter-planting measures in developing sloping economic forests can protect the ecological environment and prevent soil erosion. The positive effects of inter-planting with teas on hydrological ecosystem services should be tested extensively in other agroforestry systems in the future.

CRedit authorship contribution statement

Zhaofei Fan: Methodology. **Xia Liu:** Conceptualization. **Yiling Zheng:** Investigation. **Lei Sun:** Investigation. **Abdul Hakim Jamshidi:** Investigation. **Yun Zhu:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are grateful for the support provided by the National Natural Science Foundation of China (32071840 and 31070627), the National Key Research and Development Program of China (2018YFC0507005), the Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX241260) and Jiangsu Provincial Government Scholarship for Studying Abroad during this investigation. We would also like to thank all the supporters of this project and referees for their constructive comments.

Data availability

Data will be made available on request.

References

- Abdul-Megid, A.H., Schuman, G.E., Hart, R.H., 1987. Soil bulk density and water infiltration as affected by grazing systems. *J. Range Manag.* 40 (4), 307–309.
- Ali, H.E., Reineking, B., Münkemüller, T., 2017. Effects of plant functional traits on soil stability: intraspecific variability matters. *Plant Soil* 411 (1–2), 359–375.
- Alletto, L., Coquet, Y., 2009. Temporal and spatial variability of soil bulk density and near-saturated hydraulic conductivity under two contrasted tillage management systems. *Geoderma* 152 (1), 85–94.
- Ando, Y., Suda, K., Konishi, S., Akagi, H., 2015. Rigid plastic FE slope stability analysis combined with rain fall water infiltration. *Jpn. Geotech. Soc. Spec. Publ.* 1 (3), 23–28.
- Archer, N.A.L., Quinton, J.N., Hess, T.M., 2002. Below-ground relationships of soil texture, roots and hydraulic conductivity in two-phase mosaic vegetation in South-east Spain. *J. Arid Environ.* 52 (4), 535–553.
- Bargués Tobella, A., Reese, H., Almaw, A., Bayala, J., Malmer, A., Laudon, H., Ilstedt, U., 2014. The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semi-arid Burkina Faso. *Water Resour.* 50 (4), 3342–3354.
- Benegas, L., Ilstedt, U., Rouspard, O., Jones, J., Malmer, A., 2014. Effects of trees on infiltrability and preferential flow in two contrasting agroecosystems in Central America. *Agr. Ecosyst. Environ.* 183, 185–196.
- Blevins, R.L., Holowaychuk, N., Wilding, L.P., 1970. Micromorphology of soil fabric at the tree root-soil interface. *Soil Sci. Soc. Am. J.* 34 (3), 460–465.
- Borden, K.A., Anglaere, L.C.N., Owusu, S., Martin, A.R., Buchanan, S.W., Addo-Danso, S.D., Isaac, M.E., 2020. Soil texture moderates root functional traits in agroforestry systems across a climatic gradient. *Agr. Ecosyst. Environ.* 295, 106915.
- Broersma, K., Robertson, J.A., Chanasyk, D.S., 1995. Effects of different cropping systems on soil water properties of a Boralf soil. *Commun. Soil Sci. Plant Anal.* 26, 1795–1811.
- Bromley, J., Brouwer, J., Barker, A.P., Gaze, S.R., Valentine, C., 1997. The role of surface water redistribution in an area of patterned vegetation in a semi-arid environment, southwest Niger. *J. Hydrol.* 198 (1–4), 1–29.
- Bronick, C.J., Lal, R., 2005. Soil structure and management: a review. *Geoderma* 124 (1), 3–22.
- Burr, Hersey, Ritz, J.E., Bengough, K., Ritz, G.A., Mooney Sacha, J. K., 2020. Reorganisation of rhizosphere soil pore structure by wild plant species in compacted soils. *J. Exp. Bot.* 71 (19), 6107–6115.
- Cameron, D.R., Shaykewich, C., deJong, E., 1981. Physical aspects of soil degradation. In: *Agricultural Land – Our Disappearing Heritage – A Symposium. Proceedings of the 18th Annual Alberta Soil Science Workshop. Alberta Soil and Feed Testing Laboratory, Edmonton, Canada*, pp. 186–255.
- Cannavo, P., Sansoulet, J., Harmand, J.M., Siles, P., Dreyer, E., Vaast, P., 2011. Agroforestry associating coffee and *Inga densiflora* results in complementarity for water uptake and decreases deep drainage in Costa Rica. *Agr. Ecosyst. Environ.* 140 (1–2), 1–13.
- Chandler, K.R., Chappell, N.A., 2008. Influence of individual oak (*Quercus robur*) trees on saturated hydraulic conductivity. *For. Ecol. Manag.* 256 (5), 1222–1229.
- Chartier, M.P., Rostagno, C.M., Pazos, G.E., 2011. Effects of soil degradation on infiltration rates in grazed semi-arid rangelands of northeastern Patagonia, Argentina. *J. Arid Environ.* 75 (7), 656–661.
- Chen, S., Cao, T., Tanaka, N., Gao, T., Zhu, L., Zou, C.B., 2018. Hydrological properties of litter layers in mixed forests in Mt. Qinling. *China iForest* 11 (1), 243–250.
- Cong, W.F., Hoffland, E., Li, L., Six, J., Sun, J.H., Bao, X.G., Zhang, F.S., Van Der Werf, W., 2023. Intercropping enhances soil carbon and nitrogen. *Glob. Change Biol.* 21 (4), 1715–1726.
- Cui, Z., Wu, G.L., Huang, Z., Liu, Y., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. *J. Hydrol.* 578, 124023.
- Devitt, D.A., Smith, S.D., 2002. Root channel macropores enhance downward movement of water in a Mojave Desert ecosystem. *J. Arid Environ.* 50 (1), 99–108.
- Dexter, R.A., 2004. Soil physical quality Part I. Theory, effects of soil texture, density, and organic matter, and effects on root growth. *Geoderma* 120 (3–4), 201–214.
- Dixon, P., 2003. VEGAN, a package of R functions for community ecology. *J. Veg. Sci.* 14 (6), 927–930.
- Ebel, B.A., Moody, J.A., 2013. Rethinking infiltration in wildfire-affected soils. *Hydrol. Process* 27 (10), 1510–1514.
- Frouz, J., Livečková, M., Albrechtová, J., Chronáková, A., Cajthaml, T., Pižl, V., Hněl, L., Starý, J., Baldrian, P., Lhotáková, Z., Šimáček, H., 2013. Is the effect of trees on soil properties mediated by soil fauna? A case study from postmining sites. *For. Ecol. Manag.* 309, 87–95.
- Fu, T., Gao, H., Liang, H., Liu, J., 2021. Controlling factors of soil saturated hydraulic conductivity in Taihang Mountain Region, northern China. *Geoderma Reg.* 26, e00417.
- Galloway, A.F., Pedersen, M.J., Merry, B., Marcus, S.E., Blacker, J., Benning, L.G., Field, K.J., Knox, J.P., 2018. Xyloglucan is released by plants and promotes soil particle aggregation. *New Phytol.* 217 (3), 1128–1136.
- Ghestem, M., Veylon, G., Bernard, A., Vanel, Q., Stokes, A., 2014. Influence of plant root system morphology and architectural traits on soil shear resistance. *Plant Soil* 377 (1–2), 43–61.
- Gonzalez-Sosa, E., Braud, I., Dehotin, J., Lassabatère, L., Angulo-Jaramillo, R., Lagouy, M., Branger, F., Jacqueminet, C., Kermadi, S., Michel, K., 2010. Impact of land use on the hydraulic properties of the topsoil in a small French catchment. *Hydrol. Process* 24 (17), 2382–2399.
- Green, W.H., Ampt, G.A., 1911. Studies on Soil Physics. *J. Agric. Sci.* 4 (7–8), 1–24.
- Guo, L., Liu, Y., Wu, G.L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.Q., Tian, F.P., He, H., 2019. Preferential water flow: Influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. *J. Hydrol.* 578, 124019.
- Hao, H.X., Wei, Y.J., Cao, D.N., Guo, Z.L., Shi, Z.H., 2020. Vegetation restoration and fine roots promote soil infiltrability in heavy-textured soils. *Soil Res.* 198, 104542–104542.
- Helvey, J., Patric, J.H., 1965. Canopy and litter interception of rainfall by hardwoods of eastern United States. *Water Resour.* 1 (2), 193–206.
- Hu, W., Shao, M., Wang, Q., Fan, J., Horton, R., 2009. Temporal changes of soil hydraulic properties under different land uses. *Geoderma* 149 (3–4), 355–366.
- Huang, Z., Sun, L., Liu, Y., Liu, Y.F., López-Vicente, M., Wei, X.H., Wu, G.L., 2019. Alfalfa planting significantly improved alpine soil water infiltrability in the Qinghai-Tibetan Plateau. *Agr. Ecosyst. Environ.* 285, 106606.
- Huang, Z., Tian, F.P., Wu, G.L., Liu, Y., Dang, Z.Q., 2017. Legume grasslands promote precipitation infiltration better than gramineous grasslands in arid regions. *Land Degrad. Dev.* 28 (1), 309–316.
- Huang, L., Zhang, P., Hu, Y., Zhao, Y., 2015. Vegetation succession and soil infiltration characteristics under different aged refuse dumps at the Heidaigou opencast coal mine. *Glob. Ecol. Conserv.* 4 (C), 255–263.
- Ilstedt, U., Malmer, A., Verbeeten, E., Muriyarsa, D., 2007. The effect of afforestation on water infiltration in the tropics: a systematic review and meta-analysis. *For. Ecol. Manag.* 251 (1–2), 45–51.
- Jacka, L., Pavlásek, J., Kuráz, V., Pech, P., 2014. A comparison of three measuring methods for estimating the saturated hydraulic conductivity in the shallow subsurface layer of mountain podzols. *Geoderma* 82–88, 219–220.
- Jiang, X.J., Chen, C., Zhu, X., Zakari, S., Singh, A.K., Zhang, W., Zeng, H., Yuan, Z.Q., He, C., Yu, S., Liu, W., 2019. Use of dye infiltration experiments and HYDRUS-3D to interpret preferential flow in soil in a rubber-based agroforestry systems in Xishuangbanna, China. *Catena* 178, 120–131.
- Jiang, X.J., Liu, W., Chen, C., Liu, J., Yuan, Z.Q., Yu, X., 2018. Effects of three morphometric features of roots on soil water flow behavior in three sites in China. *Geoderma* 320, 161–171.
- Jørgensen, P.R., Hoffmann, M., Kistrup, J.P., Bryde, C., Bossi, R., Villholth, K.G., 2002. Preferential flow and pesticide transport in a clay-rich till: field, laboratory, and modeling analysis. *Water Resour.* 38 (11), 1246–1261.
- Jotisankasa, A., Sirirattanachai, M.T., 2017. Effects of grass roots on soil-water retention curve and permeability. *Can. Geotech. J.* 54 (11), 1612–1622.
- Kan, X., Cheng, J., Hou, F., 2020. Response of preferential soil flow to different infiltration rates and vegetation types in the karst region of southwest China. *Water* 12 (6), 1778.
- Lacombe, S., Bradley, R.L., Hamel, C., Beaulieu, C., 2009. Do tree-based intercropping systems increase the diversity and stability of soil microbial communities? *Agr. Ecosyst. Environ.* 131 (1), 25–31.
- Lehmann, J., da Silva Cravo, M., Zech, W., 2001. Organic matter stabilisation in a Xanthic Ferralsol of the central Amazon as affected by single trees: chemical characterization of density, aggregate and particle size fractions. *Geoderma* 99 (1), 147–168.

- Leung, A.K., Garg, A., Coo, J.L., Ng, C.W.W., Hau, B.C.H., 2015. Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrol. Process* 29 (15), 3342–3354.
- Li, Y., Li, B., Zhang, X., Chen, J.J., Zhan, F.D., Guo, X.H., Zu, Y.Q., 2015. Differential water and soil conservation capacity and associated processes in four forest ecosystems in Dianchi Watershed, Yunnan Province, China. *J. Soil Water Conserv* 70 (3), 198–206.
- Li, N., Zhang, Y., Wang, T., Li, J., Yang, J., Luo, M., 2022. Have anthropogenic factors mitigated or intensified soil erosion over the past three decades in South China? *J. Environ. Manag.* 302 (PB), 114093.
- Liu, Y., Cui, Z., Huang, Z., López-Vicente, M., Wu, G.L., 2019. Influence of soil moisture and plant roots on the soil infiltration capacity at different stages in arid grasslands of China. *Catena* 182, 104147.
- Liu, Y., Guo, L., Huang, Z., López-Vicente, M., Wu, G.L., 2020. Root morphological characteristics and soil water infiltration capacity in semi-arid artificial grassland soils. *Agric. Water Manag.* 235, 106153.
- Liu, Y., Miao, H.T., Chang, X., Wu, G.L., 2019. Higher species diversity improves soil water infiltration capacity by increasing soil organic matter content in semiarid grasslands. *Land Degrad. Dev.* 30 (13), 1599–1606.
- Liu, L.C., Wu, Z.Y., Liu, J.Q., 2015. Effects of land use on spatial patterns of soil properties in a rocky mountain area of Northern China. *Arab. J. Geosci.* 8 (2), 1181–1194.
- Liu, W., Zhu, C., Wu, J., Chen, C., 2016. Are rubber-based agroforestry systems effective in controlling rain splash erosion? *Catena* 147, 16–24.
- Marquart, A., Eldridge, D.J., Geissler, K., Lobas, C., Blaum, N., 2020. Interconnected effects of shrubs, invertebrate-derived macropores and soil texture on water infiltration in a semi-arid savanna rangeland. *Land Degrad. Dev.* 31 (16), 2307–2318.
- Mo, J.M., Brown, S., Peng, S.L., Kong, G.H., Fang, Y.T., 2005. Effects of human impacts on fine roots and soil organic matter of a pine forest in subtropical China. *Acta Ecol. Sin.* 25 (3), 491–499 (In Chinese).
- Molina, A., Govers, G., Vanacker, V., Poesen, J., Zeelmaekers, E., Cisneros, F., 2007. Runoff generation in a degraded Andean ecosystem: interaction of vegetation cover and land use. *Catena* 71 (2), 357–370.
- Neris, J., Jiménez, C., Fuentes, J., Morillas, G., Tejedor, M., 2012. Vegetation and land use effects on soil properties and water infiltration of Andisols in Tenerife (Canary Islands, Spain). *Catena* 98, 55–62.
- Ng, C.W.W., Woon, K.X., Leung, A.K., Chu, L.M., 2013. Experimental investigation of induced suction distributions in a grass-covered soil. *Ecol. Eng.* 52, 219–223.
- Niu, C.Y., Shi, Y.F., Fan, G.Q., Ye, Y.L., Zhang, J.S., Sang, Y.Q., Zhang, Z.H., 2023. Effect of Rock Fragments on Soil Water and Nutrient: a Case Study in Rocky Mountain Area of North China. *J. Soil Sci. Plant Nutr.* 24 (1), 716–731.
- Pan, C., Ma, L., Wainwright, J., Shanguan, Z., 2016. Overland flow resistances on varying slope gradients and partitioning on grassed slopes under simulated rainfall. *Water Resour.* 52 (4), 2490–2512.
- Persson, H., 1980. Spatial distribution of fine-root growth, mortality and decomposition in a young Scots pine stand in Central Sweden. *Oikos* 34 (1), 77–87.
- Poulenard, J., Podwojewski, P., Janeau, J.L., Collinet, J., 2001. Runoff and soil erosion under rainfall simulation of Andisols from the Ecuadorian Páramo: effect of tillage and burning. *Catena* 45 (3), 185–207.
- Rodríguez Rodríguez, A., Guerra, J.A., Gorrín, S.P., Arbelo, C.D., Mora, J.L., 2002. Aggregates stability and water erosion in Andosols of the Canary Islands. *Land Degrad. Dev.* 13 (6), 515–523.
- Sanchez, G., 2013. Editions ed. PLS Path Modeling with R, Trowchez. Trowchez Editions, Berkeley.
- Sharma, N.K., Singh, R.J., Mandal, D., Kumar, A., Alam, N.M., Keesstra, S., 2017. Increasing farmer's income and reducing soil erosion using intercropping in rainfed maize-wheat rotation of Himalaya, India. *Agr. Ecosyst. Environ.* 247, 43–53.
- Shi, S., Zhao, F., Ren, X., Meng, Z., Dang, X., Wu, X., 2022. Soil infiltration properties are affected by typical plant communities in a semi-arid desert grassland in China. *Water* 14 (20), 3301.
- Shukla, M.K., Lal, R., Owens, L.B., Unkefer, P., 2003. Land use and management impacts on structure and infiltration characteristics of soils in the North Appalachian region of Ohio. *Soil Sci.* 168 (3), 167–177.
- Six, J., Bossuyt, H., Degryze, S., Denef, K., 2004. A history of research on the link between (micro) aggregates, soil biota, and soil organic matter dynamics. *Soil Res.* 79 (1), 7–31.
- Sun, W., Shao, Q., Liu, J., Zhai, J., 2014. Assessing the effects of land use and topography on soil erosion on the Loess Plateau in China. *Catena* 121, 151–163.
- Sun, J., Yu, X., Wang, H., Jia, G., Zhao, Y., Tu, Z., Deng, W., Jia, J., Chen, J., 2018. Effects of forest structure on hydrological processes in China. *J. Hydrol.* 561, 187–199.
- Thompson, S.E., Harman, C.J., Heine, P., Katul, G.G., 2010. Vegetation-infiltration relationships across climatic and soil type gradients. *J. Geophys. Res.-Biogeog.* 115 (G2) (n/a - n/a).
- Tollner, E.W., Calvert, G.V., Langdale, G., 1990. Animal trampling effects on soil physical properties of two Southeastern U.S. Ultisols. *Agr. Ecosyst. Environ.* 33 (1), 75–87.
- Tu, Z., Chen, S., Ruan, D., Chen, Z., Huang, Y., Chen, J., 2022. Differential hydrological properties of forest litter layers in artificial afforestation of eroded areas of latosol in China. *Sustainability* 14 (22), 14869.
- Vergani, C., Graf, F., 2016. Soil permeability, aggregate stability and root growth: a pot experiment from a soil bioengineering perspective. *Ecology* 9 (5), 830–842.
- Wang, F., Chen, H., Lian, J., Fu, Z., Nie, Y., 2018. Preferential flow in different soil architectures of a small karst catchment. *Vadose Zone J.* 17 (1), 1539–1663.
- Wang, R., Dong, Z., Zhou, Z., Wang, N., Xue, Z., Cao, L., 2020. Effect of vegetation patchiness on the subsurface water distribution in abandoned farmland of the Loess Plateau, China. *Sci. Total Environ.* 746, 141416.
- Wang, B., Liu, J., Li, Z., Morreale, S.J., Schneider, R.L., Xu, D., Lin, X., 2023. The contributions of root morphological characteristics and soil property to soil infiltration in a reseeded desert steppe. *Catena* 225, 141416.
- Wang, L., Wang, Q., Wei, S., Shao, M.A., Li, Y., 2008. Soil desiccation for Loess soils on natural and regrown areas. *For. Ecol. Manag.* 255 (7), 2467–2477.
- Wang, H., Zhang, G., Liu, F., Geng, R., Wang, L., 2017. Effects of biological crust coverage on soil hydraulic properties for the Loess Plateau of China. *Hydrol. Process* 31 (19), 3396–3406.
- Wei, X., Bi, H., Liang, W., Hou, G., Kong, L., Zhou, Q., 2018. Relationship between soil characteristics and stand structure of *Robinia pseudoacacia* L. and *Pinus tabulaeformis* Carr. Mixed Plantations in the Caijiachuan Watershed: An Application of Structural Equation Modeling. *Forests* 9 (3), 124.
- Whalley, W.R., Leeds-Harrison, P.B., Leech, P.K., Riseley, B., Bird, N.R.A., 2004. The hydraulic properties of soil at root-soil interface. *Soil Sci.* 169 (2), 90–99.
- Wu, G.L., Cui, Z., Huang, Z., 2021. Contribution of root decay process on soil infiltration capacity and soil water replenishment of planted forestland in semi-arid regions. *Geoderma* 404, 115289.
- Wu, J.E., Liu, W.J., Chen, C.F., 2016. Can intercropping with the world's three major beverage plants help improve the water use of rubber trees? *J. Appl. Ecol.* 53 (6), 1787–1799.
- Wu, G.L., Yang, Z., Cui, Z., Liu, Y., Fang, N.F., Shi, Z.H., 2016. Mixed artificial grasslands with more roots improved mine soil infiltration capacity. *J. Hydrol.* 535, 54–60.
- Xu, X., Lewis, C., Liu, W., Albertson, J.D., Kiely, G., 2012. Analysis of single-ring infiltrometer data for soil hydraulic properties estimation: comparison of BEST and Wu method. *Agr. Water Manag.* 107 (C), 34–41.
- Yimer, F., Messing, I., Ledin, S., Abdelkadir, A., 2008. Effects of different land use types on infiltration capacity in a catchment in the highlands of Ethiopia. *Soil Use Manag.* 24 (4), 344–349.
- Zhang, W., Wang, K., Chen, H., He, X., Zhang, J., 2012. Ancillary information improves kriging on soil organic matter content data for a typical karst peak cluster depression landscape. *J. Sci. Food Agric.* 92 (5), 1094–1102.
- Zhang, G.H., Xie, Z.F., 2019. Soil surface roughness decay under different topographic conditions. *Soil Res.* 187, 92–101.
- Zhao, X., Wu, P., Gao, X., Tian, L., Li, H., 2014. Changes of soil hydraulic properties under early stage natural vegetation recovering on the loess plateau of China. *Catena* 113 (1), 386–391.

- Zhou, Q., Keith, D.M., Zhou, X., Cai, M., Cui, X., Wei, X., Luo, Y., 2018. Comparing the water-holding characteristics of broadleaved, coniferous, and mixed forest litter layers in a Karst Region. *Mt. Res. Dev.* 38 (3), 220–229.
- Zhu, X., Chen, C., Wu, J., Yang, J., Zhang, W., Zou, X., Liu, W., Jiang, X., 2019. Can intercroops improve soil water infiltrability and preferential flow in rubber-based agroforestry system? *Soil Res.* 191, 327–339.
- Zhu, X.A., Liu, W.J., Jiang, X.J., Wang, P.Y., Li, W.X., 2018. Effects of land-use changes on runoff and sediment yield: implications for soil conservation and forest management in Xishuangbanna, SW China. *Land Degrad. Dev.* 29 (9), 2962–2974.
- Zhu, P.Z., Zhang, G.H., Wang, H.X., Xing, S.K., 2020. Soil infiltration properties affected by typical plant communities on steep gully slopes on the Loess Plateau of China. *J. hydrol.* 590, 125535.